

Mega XL — custom upgrades & build notes

Upgrades to a Kickstarter v1.0 Mega XL CNC router.

This documents the changes made to a first-generation (Kickstarter v1.0) Mega XL: a taller, stiffer Z, closed-loop motors, all-proximity homing, and a complete electronics rebuild around a grblHAL controller in a custom enclosure with a proper 48 V power system, air assist, and probing.

At a glance

Subsystem	Stock (v1.0)	Upgraded
Vertical clearance	stock	+3" via Mark Smith riser kit
Z assembly	stock Z	Mark Smith Z gantry (only V-wheels + X motor mount reused)
Spindle	DeWalt DWP611 (1¼ HP router)	2.2 kW (3 HP) Ø80 mm water-cooled, ER20, 24k RPM, VFD-driven
Stepper motors	generic open-loop steppers	all-in-one closed-loop steppers (driver integrated in motor)
Limit / homing	mechanical limit switches	proximity sensors on every axis
Probing	—	touch plate + 3D probe in parallel; grounded spindle (no tool clip)
Controller	8-bit Arduino board	grblHAL Teensy 4.x (USB + Ethernet)
Electronics enclosure	cramped stock steel box	custom 12×16×6" aluminum / plexiglass
Power	stock	48 V Meanwell system, E-stop, buck-derived 12 V / 5 V rails
Air assist	none	Mist-relay 12 V solenoid → 30 PSI nozzle at the cut
Drag chains	stock 1.5	Y axis upsized to 2" to fit the full cable bundle
Dust collection	—	4" swivel + 5' flex to a spindle dust shoe, on an overhead reel

Frame & mechanical

- **Mark Smith riser kit** — adds **3" of additional vertical (Z) clearance**, allowing taller stock and longer tooling.
- **Mark Smith Z gantry** — a **complete replacement of the Z assembly**. The only stock parts carried over are the **four V-wheels** and the **X stepper motor mount**; everything else is new.

Spindle & cooling

The stock **DeWalt DWP611** (1¼ HP trim router) was replaced with a **2.2 kW (3 HP) water-cooled spindle** kit:

- **Spindle** — Ø80 mm body, 220 V, water-cooled, **ER20 collet**, up to **24,000 RPM at 400 Hz**.
- **VFD** — 2.2 kW (3 HP) inverter, fed from the 220 V circuit through the E-stop and commanded by the breakout board's **0–10 V PWM** speed signal (see Power distribution).
- **Mount** — 80 mm spindle clamp on the new Z gantry.
- **Braking resistor** — a 150 W resistor on the VFD dissipates regen energy for a fast stop (spindle down in **under 6 s**).

Coolant loop. A pump **submerged in a 1-gallon reservoir** circulates coolant through the spindle via **two 8 mm tubes** (supply and return): *reservoir* → *pump* → *spindle* → *radiator* → *reservoir*. The radiator is cooled by the 12 V fan (see the power tree). The pump is **interlocked to the spindle** — a **VFD digital output** drives the **NO contact of a 110 V relay** the pump plugs into — so it runs whenever the spindle does and the spindle can't be run dry. (Separate from the breakout board's Mist relay, which handles air assist.) The coolant tubes, the air-assist line, and dust extraction all reach the spindle through the overhead umbilical (below).

Motors

The generic steppers that shipped with the Mega XL were replaced with **all-in-one closed-loop stepper motors** — each motor has its stepper driver built into the motor body. This removes the separate driver boards, adds closed-loop position feedback (lost-step detection / alarm), and simplifies wiring to a single cable per motor.

- **Individual enable** per motor — each stepper's **EN** line is driven separately.
- The motors' **AL (alarm) outputs are wire-OR'd together** and fed to the controller's **ST0 input**, so a stall / lost-step fault on any motor signals the controller.

All four axes are closed-loop. The machine is a **dual-Y gantry** — the two Y gantries are driven by ganged motors (**Y + A**), one per side.

Axis	Holding torque	Drives
Y + A	3.2 N·m each	the two Y gantries (ganged)
X	2.0 N·m	spindle carriage along the cross-beam (rack & pinion)
Z	1.2 N·m	Z

The breakout board also provides a **5th axis (B)**. No rotary is fitted yet, but its **step/dir/enable is wired through to the extension panel** so one can be added later by plugging it in — see [Back-of-enclosure harness & connectors](#).

All motors have an **8 mm output shaft with a flat**, with pinions/couplings retained by a grub screw on the flat. *Build note*: the stock Mega XL pinions were 6 mm-bore and were **reamed to 8 mm** at a machine shop — this thinned the set-screw boss and reduced grub-screw thread engagement. See the [rack-and-pinion maintenance](#) appendix.

Homing & limit sensing

The Mark Smith Z gantry uses a **proximity sensor** as its limit switch. To keep one sensing scheme across the machine, **all remaining stock mechanical limit switches were replaced with the same style of proximity sensor**, so every axis homes on identical, non-contact hardware.

Probing

- The **touch-plate input is wired in parallel with the primary probe input**, alongside the **3D mechanical probe** — either device can trigger the same probe input.
- The **spindle mount is grounded**, so the tool itself is at ground through the spindle. That completes the touch-plate circuit on contact — **no alligator clip on the tool** is required.

Controller & electronics enclosure

- Discarded the **cramped stock steel electronics enclosure** and the **stock 8-bit Arduino controller board**.
- Control is now a **grblHAL Teensy 4.x** controller on a breakout board (32-bit).
- Built a **custom 12" × 16" × 6" enclosure**:
 - Frame: **2020 aluminum extrusion**.
 - Five sides: **0.0625" (1/16") aluminum sheet**.
 - Top: **1/8" clear plexiglass** (visibility into the electronics).

Connectivity

The Teensy 4.x supports Ethernet, and the breakout board has the **Ethernet option installed**. Both links to the Teensy are brought out as **panel patch-through connectors on the side of the enclosure**: a **female USB-A** jack and a **female Ethernet (RJ45)** jack.

Power distribution

A single **220 V, 20 A branch circuit** feeds the machine through a **front-panel emergency-stop button** (the E-stop breaks the 220 V input, so it drops both the PSU and the spindle drive). The 220 V powers an **800 W 48 V Meanwell power supply** and the **VFD** (spindle drive). Everything else is derived from the 48 V rail through two buck converters.

220 V, 20 A → front-panel E-stop →

- └ 48 V Meanwell power supply
- └ VFD (spindle drive)

48 V (PSU output) →

- └ 4 × stepper motors
- └ 12 V / 3 A buck converter
- └ 5 V / 3 A buck converter

12 V (buck) →

- └ 4 × proximity limit sensors
- └ 12 V fan cooling the spindle coolant loop (runs while the spindle runs)
- └ grblHAL Teensy 4.x breakout board, 12 V input — the board uses 12 V to generate the **0–10 V PWM spindle-speed signal** to the VFD

5 V (buck) →

- └ breakout board 5 V input — board logic power, **separate from the Teensy** (the Teensy is powered over USB)
- └ 3D probe

Relay outputs & air assist

The breakout board provides **5 relay outputs**, each **jumper-selectable for 5 V or 12 V**. Only the **Mist** output is in use, driving the air-assist nozzle:

Mist relay → 12 V **NC** solenoid valve → regulator @ 30 PSI → 1/4" tube (up the umbilical) → needle valve at top of Z gantry → 1/8" copper airline → swivel nozzle aimed where the tool meets the stock

Connectors & terminations

Ferrules are crimped on all terminal-block connections — every stranded conductor landing on a terminal block is ferruled for a clean, captured, reliable termination.

Enclosure pass-throughs use **Molex Mini-Fit Jr panel-mount connectors**. The choice was deliberate: with roughly 60 terminations to make, the builder had far more confidence producing **~60 quality crimps** than ~60 quality solder joints on aircraft-style connectors.

Cable & wiring schedule

All runs use shielded cable.

Cable	Run	Conductors
18/6 shielded	enclosure → each stepper motor	6 (assignment depends on motor type — see below)
22/3 shielded	enclosure → Y-axis proximity switches	12V , Sig , GND
22/6 shielded	enclosure → X & Z proximity sensors + 3D probe (shared)	12V , GND , XLim , ZLim , P , 5V
12/4 shielded	VFD → spindle (3-phase motor power)	U , V , W , GND
22/4 shielded	breakout → VFD (control)	0-10V , run, GND, spare

18/6 stepper cable — conductor assignment

Conductor	Closed-loop stepper	Discrete (open-loop) stepper
1	48V	A+
2	GND	A-
3	DIR	B+
4	PUL	B-
5	EN	EN
6	AL	AL

All four axes on this build are closed-loop, so they all use the left-column assignment (48V / GND / DIR / PUL / EN / AL). DIR / PUL are direction/step to the integrated drives; EN = enable, AL = alarm. The discrete A± / B± column is kept for reference only — the same 18/6 cable would serve an open-loop motor if one were ever fitted. At the enclosure these device cables collapse onto a small set of keyed panel connectors — see [Back-of-enclosure harness & connectors](#).

Drag chains (cable carriers)

Each axis's drag chain carries every cable that has to continue past it, so loading is cumulative toward the base of the motion chain. **Y is the fullest** — everything routes through the Y carrier first to reach the gantry, then the X carrier feeds the spindle carriage, while the opposite gantry rail (**A**) carries only its own stepper. The two limit-sensor drops that don't need to flex (A and Y) **begin after** their drag chains rather than riding through them.

Drag chain	Cables carried
A axis (opposite gantry rail)	1 × 18/6 (A stepper) <i>A limit (22/3) picks up after the chain</i>
X axis (spindle carriage)	1 × 12/4 (spindle power) 2 × 18/6 (X & Z steppers) 1 × 22/6 (X & Z limits + 3D probe)
Y axis (gantry — carries everything)	1 × 12/4 (spindle power) 3 × 18/6 (X, Z & Y steppers) 1 × 22/6 (X & Z limits + 3D probe) <i>Y limit (22/3) picks up after the chain</i>

That Y bundle — one 12/4, three 18/6, and one 22/6 — would not fit the stock **1.5** drag chain, so the Y axis was moved up to the **next size larger (2")** carrier.

Back-of-enclosure harness & connectors

The control enclosure lives in a **cabinet drawer**. The only practical way to reach the back is to pull the drawer fully forward and connect the harness **by feel** — so the wiring was deliberately collapsed onto a **few keyed Mini-Fit Jr sockets** rather than many small connectors. Every socket is **1×6 (single-row)**: a linear row of pins is far easier to count and seat blind than a 2×3 grid.

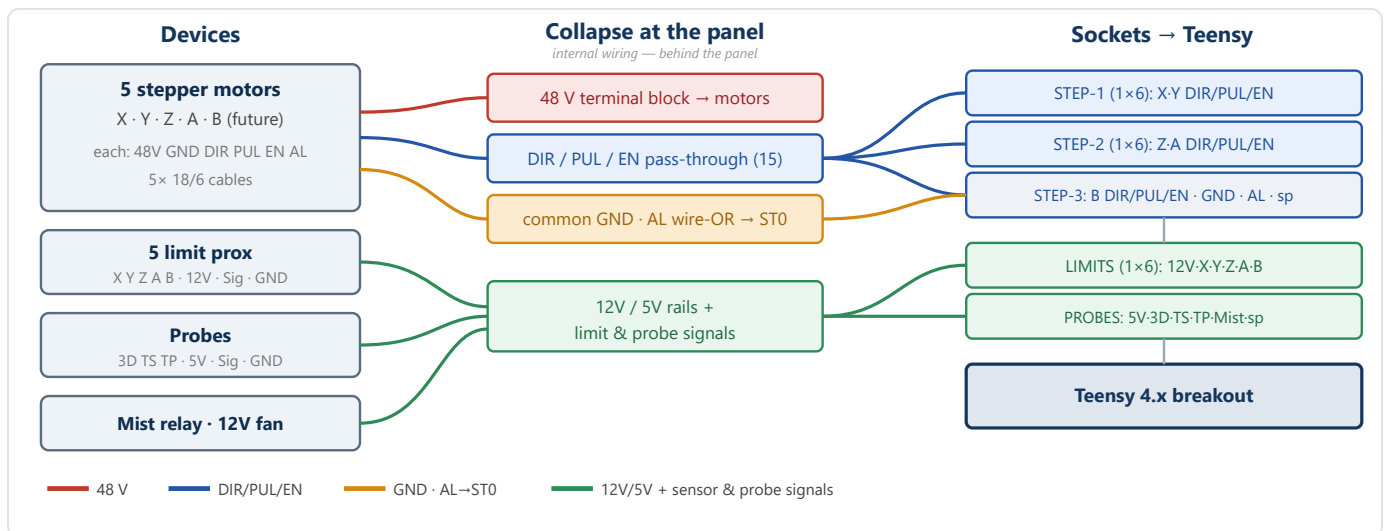
The collapse. Two facts shrink the conductor count dramatically:

- **GND is common** (tied to the supply's V–), so every device ground is the same node — one shared ground instead of one per device.
- Each closed-loop motor only needs **DIR, PUL, EN** unique — its 48 V is its own rail, GND is shared, and the five **AL** **alarms wire-OR into a single ST0**.

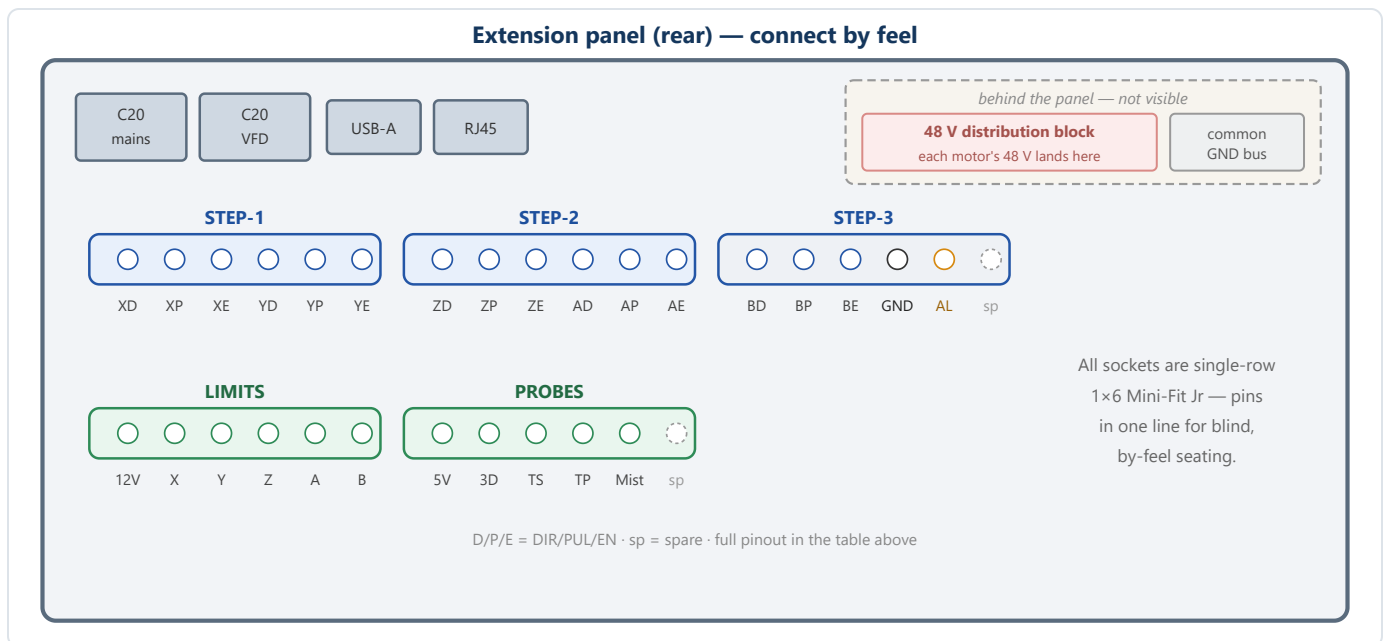
That takes the 5 stepper channels (X, Y, Z, A, and the provisioned B) from 30 conductors down to **5×3 = 15 + shared GND + OR'd AL = 17**, which fit **three 1×6 sockets** (one spare pin). Each motor's **48 V runs separately to a distribution terminal block** at the panel — not through these signal sockets. The sensors and probes fill **two more 1×6 sockets**. Net: **five cables / five sockets** to the controller (three 18/6 for the steppers, two 22/6 for the low-current sensor lines), plus the 48 V block and the common-ground tie. The **48 V terminal block and common-ground bus sit behind the panel** — internal wiring, not on the connect-by-feel face. The spindle's **VFD has its own C20 220 V inlet** on the panel and a **22/4 control cable** from the breakout (0–10 V speed + run).

Socket (1×6)	Cable	1	2	3	4	5	6
Stepper 1	18/6	X DIR	X PUL	X EN	Y DIR	Y PUL	Y EN
Stepper 2	18/6	Z DIR	Z PUL	Z EN	A DIR	A PUL	A EN
Stepper 3	18/6	B DIR	B PUL	B EN	GND	AL	spare
Limits	22/6	12V	X	Y	Z	A	B
Probes	22/6	5V	3D	TS	TP	Mist	spare

B rows are the provisioned 5th-axis / rotary lines. **3D** = 3D probe, **TS** = toolsetter, **TP** = touch plate. The 12 V fan is fed from the panel, not the probes socket; each motor's 48 V and the common ground are distributed separately.



Harness collapse: many device wires → shared GND, wire-OR'd alarm and a 48 V block → five 1×6 sockets into the controller.



Back-panel layout: the five 1×6 sockets plus the 48 V block, common-ground bus and the USB/Ethernet/C20 pass-throughs.

Overhead umbilical & dust collection

The spindle's services arrive from **above** rather than through a drag chain, on a **ceiling-mounted spring retraction reel** that takes up slack as the gantry moves.

- **Umbilical:** the two 8 mm coolant tubes and a 1/4" compressed-air tube run from the top of the Z gantry into a **1" cable sleeve**.

- **Retraction reel:** the sleeve hangs from a spring-loaded retraction reel on the ceiling above the machine. A **3D-printed cradle** holds the sleeve and clips to the reel hook, so the reel's pull-wire can't wrap and choke the sleeve while still allowing free movement.
- **Routing:** the sleeve terminates at the **back of the CNC**, where the coolant tubes join the pump outlet and the radiator inlet (radiator outlet returns to the reservoir). The air tube comes from the pressure-regulator outlet, whose inlet is fed by the 12 V NC Mist solenoid.
- **Dust collection:** alongside the umbilical, a **4" duct** ends in a **4" swivel** feeding **5' of 4" flex hose** to a **dust shoe** on the spindle.

Rack-and-pinion drive — setup & maintenance

X (and the Y/A gantry drives) run on rack and pinion. After ~10 hours the X pinion crept **left** under load. A closed-loop motor can't see slip *downstream* of its shaft encoder, so a pinion loosening on the shaft is invisible until a fixed reference (the toolsetter at G59.3) exposes it. With the bores reamed 6→8 mm the set-screw boss is thin and grub-screw grip is the weak point; X slips first because it takes the most direct cutting side-load, but the other reamed pinions carry the same risk.

Pinion-to-shaft — stop the slip

1. Pull the pinion; confirm the grub screw seats on the shaft **flat** (not a round section), and check whether the screw marred the flat or only kissed it.
2. **Degrease** shaft, bore, and screw threads — oil defeats both grip and threadlocker.
3. Use the **longest grub screw** the thinned boss will take, on the flat; add a second screw if there's room.
4. **Threadlocker:** blue (242/243) is normally enough; with the reduced thread engagement here, **red (271)** is justified as insurance against back-out — just note removal then needs heat (~250 °C, easy on a bare pinion).
5. Torque to screw size (M3 ~1.0–1.4, M4 ~2.5–3, M5 ~5–6 N·m); snug, don't strip the hex. Let red cure ~24 h before heavy cuts.

Pinion-to-rack mesh & backlash

1. Loosen the motor mount so the pinion can move toward / away from the rack.
2. Shim **0.05–0.10 mm** (feeler gauge, or ~0.1 mm folded paper) between a pinion tooth and the rack, push the pinion in against the shim, tighten the mount, remove the shim.
3. Check the mesh over the **full travel** — it should stay lightly meshed everywhere. Tight at one end + slack at the other means the rack isn't parallel to the axis; realign the rack first.
4. Confirm full tooth-width engagement (pinion not riding a tooth edge).

Verify

1. By hand: smooth full travel, minimal lash; rock the gantry and feel for play at the pinion.
2. Dial indicator: commanded vs actual on small moves; reverse direction to read backlash (a few thou).
3. Re-run a heavy back-and-forth / cut loop with paint-pen **witness marks** (motor shaft ↔ pinion, and a pinion tooth ↔ a rack tooth) — no separation = fixed.
4. Confirm the `AL → ST0` alarm actually halts grbIHAL (force a stall) so a real fault can't pass silently.

If it still creeps

A single grub screw — especially in a reamed, thin-wall boss — is marginal for rack-and-pinion loads. Durable fixes, in order: **proper 8 mm-bore pinions** (full set-screw boss) → **clamp / split-hub pinion** (grips the whole shaft, no set screw) → **keyed shaft**. Given the reamed pinions, the **clamp-hub style is the recommended upgrade** — it ends set-screw slip for good.

Sourcing a replacement pinion

Match the rack on three things, plus the bore:

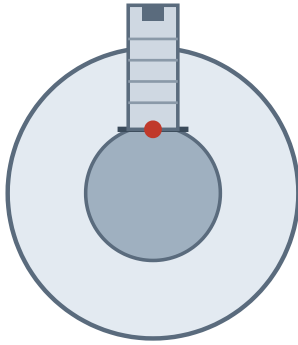
- **Module** (tooth size) — `module = tooth pitch ÷ π` (pitch 3.14 mm = Mod 1, 4.71 mm = Mod 1.5, 6.28 mm = Mod 2). Must match the rack exactly.
- **Pressure angle** — almost always **20°**.
- **Tooth count** — match the original to keep steps/mm, or pick a new count and recalibrate `$100` (X steps/mm) in grbIHAL.
- **Bore** — 8 mm, ideally a **clamp or keyed** bore, not a bare set-screw boss.

Sources that let you specify *module + 20° + 8 mm clamp/keyed bore*: **KHK / QTC** (khkgears.us, qtcgears.com — best for CNC rack & pinion), **SDP/SI** (sdp-si.com), **Misumi** (configurable bore & keyway), and **McMaster-Carr** (quick metric-module steel gears). Budget CNC rack-and-pinion sets are on Amazon / AliExpress (Mod 1/1.5, 20T, often 8 mm bore) with variable quality.

Clamp hub vs set-screw hub

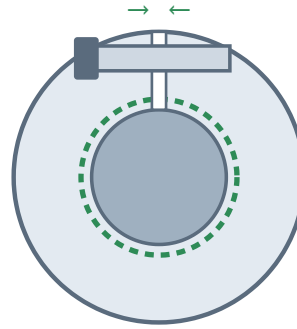
A **clamp (split) hub** grips like a clamp-style shaft collar: a slit is sawn through the bore and a screw bridges it; tightening the screw squeezes the slit closed so the **whole bore clamps the shaft 360°** — far more holding torque, no marring, and no dependence on a thin set-screw boss (exactly the reamed-pinion weakness).

Set-screw bore



point load on a flat
reamed boss = few threads = creeps

Clamp bore



slit + screw close the bore
grips the shaft 360° — no flat needed

Two routes to a clamp mount: a pinion with an **integral clamp bore** (specify it when ordering from KHK / Misumi / SDP), or a **keyless locking bushing** (Trantorque GT, Fenner B-LOC, ...) dropped inside a larger-bore pinion — the bushing expands to grip both the shaft and the bore, letting you use almost any pinion of the right module on the 8 mm shaft. The latter is handy here, since matching bore *and* gearing was the sticking point.

Bill of materials

Group	Item	Spec / source
Machine	Mega V XL (Kickstarter edition)	base machine
Frame	Mark Smith riser kit	+3" Z clearance
Frame	Mark Smith Z gantry assembly	full Z replacement
Control	Teensy 4.1 + Teensy 4.x breakout board	grblHAL, Ethernet option
I/O	USB-A pass-through, panel mount	
I/O	Ethernet pass-through, panel mount	
I/O	E-stop button	breaks 220 V input
Power	Meanwell 48 V power supply	800 W
Power	C20 panel-mount inlet (mains)	220 V mains in
Power	C20 panel-mount inlet (VFD)	220 V to the spindle VFD
Power	Power relay accessory	110 V pump relay — Amazon B07Z6R1KP3
Spindle	2.2 kW 80 mm water-cooled spindle + 2.2 kW VFD kit	ER20, 24k RPM — Amazon B073TRP92V
Spindle	Braking resistor	150 W (<6 s stop)
Cooling	Submersible pump	110 V
Cooling	Coolant reservoir	1 gal container (holds the pump)
Cooling	Radiator with integrated 12 V fan(s)	
Cooling	8 mm clear tubing	25' (coolant lines)
Cooling	Coolant mix	1 qt antifreeze + ½ gal distilled water + a few drops of bleach
Motion	Z stepper (closed-loop)	1.2 N·m
Motion	X stepper (closed-loop)	2.0 N·m
Motion	Y + A steppers (closed-loop)	3.2 N·m each (×2)
Motion	Drag chain	2" (Y-axis upsize)
Sensing	Proximity sensors	×4 (limits / homing)

Group	Item	Spec / source
Sensing	3D mechanical probe	
Sensing	Toolsetter probe	
Air assist	1/4" black tubing	10'
Air assist	Needle valve	
Air assist	Copper tubing	12"
Air assist	Swivel nozzle	1/8"
Air assist	Solenoid valve	12 V, NC
Air assist	Pressure regulator	set 30 PSI
Air assist	3D-printed nozzle bracket	bolts to Z; aims the air stream at the tool tip
Cabling	Shielded cable	22/6, 22/4, 22/3, 18/6, 12/4
Connectors	Mini-Fit Jr 1×6 connectors	5 mated pairs (5 panel headers + 5 cable receptacles)
Connectors	Mini-Fit Jr crimp terminals	~60 (≈30 pins + 30 sockets)
Connectors	48 V distribution terminal block	at the extension panel (each motor's 48 V)

Not yet itemized (described in the build above but missing from this BOM):

- **Enclosure** — 2020 extrusion, 1/16" aluminum sheet (5 sides), 1/8" plexiglass top.
- **Buck converters** — 12 V 3 A and 5 V 3 A.
- **Ferrules** for the terminal-block connections.
- **Touch plate** (probe input in parallel with the 3D probe).
- **Dust collection** — 4" duct, 4" swivel, 5' of 4" flex hose, dust shoe.
- **Overhead service rig** — spring retraction reel, 1" cable sleeve, 3D-printed sleeve cradle.